

The Solar System - A Short Guide

Introduction

The Solar System consists of the Sun and everything that orbits around it, including eight planets, their moons, dwarf planets, asteroids, comets, and meteoroids. It formed about 4.6 billion years ago from the gravitational collapse of a giant interstellar molecular cloud.

The Sun

The Sun is a nearly perfect sphere of hot plasma at the center of the Solar System. It contains 99.86 percent of the total mass of the Solar System. The Sun's energy comes from nuclear fusion, where hydrogen atoms combine to form helium in its core, releasing enormous amounts of energy as light and heat.

Mercury

Mercury is the smallest planet and the closest to the Sun. A year on Mercury lasts only 88 Earth days. It has no atmosphere to retain heat, so temperatures swing from extremely hot during the day to extremely cold at night.

Venus

Venus is the second planet from the Sun and is often called Earth's twin because of its similar size. However, Venus has a thick, toxic atmosphere made mostly of carbon dioxide, causing a runaway greenhouse effect that makes it the hottest planet in the Solar System.

Earth

Earth is the third planet from the Sun and the only known planet to support life. About 71 percent of its surface is covered in water. Earth's atmosphere is mostly nitrogen and oxygen, and it has one natural satellite, the Moon.

Mars

Mars is known as the Red Planet because of the iron oxide, or rust, on its surface. It has the largest volcano in the Solar System, Olympus Mons, and evidence suggests it once had liquid water on its surface billions of years ago.

Jupiter

Jupiter is the largest planet in the Solar System, more massive than all the other planets combined. It is a gas giant made mostly of hydrogen and helium, and is famous for its Great Red Spot, a giant storm that has raged for hundreds of years. Jupiter has at least 95 known moons, including the four large Galilean moons.

Saturn

Saturn is famous for its spectacular ring system, made mostly of ice particles with some rocky debris and dust. It is a gas giant like Jupiter and has dozens of moons, the largest of which, Titan, has a thick atmosphere and lakes of liquid methane.

Uranus

Uranus is an ice giant with a unique feature: it rotates on its side, with an axial tilt of about 98 degrees. This means its poles experience 42 years of continuous sunlight followed by 42 years of darkness. Uranus has a faint ring system and 27 known moons.

Neptune

Neptune is the farthest known planet from the Sun and is also an ice giant. It has the strongest winds in the Solar System, reaching speeds of over 2,000 kilometers per hour. Neptune was the first planet located through mathematical prediction rather than direct observation.

Dwarf Planets and Beyond

Beyond Neptune lies the Kuiper Belt, home to dwarf planets like Pluto, which was reclassified from a full planet in 2006. Other notable dwarf planets include Eris, Haumea, and Makemake. Even farther out lies the Oort Cloud, a theoretical shell of icy objects believed to be the source of long-period comets.